

Documentation

Overview

This project implements a simple AI-powered financial chatbot prototype. The chatbot uses rule-based logic to respond to predefined financial questions based on analyzed company financial data.

How It Works

1. The user enters a financial question.
2. The chatbot converts the question to lowercase.
3. The chatbot searches a dictionary containing predefined questions and responses.
4. If a match is found, the corresponding financial insight is returned.
5. If no match is found, a default response is displayed.

Supported Queries

- What is the total revenue?
- What is the net income?
- How has net income changed over the last year?
- What is the revenue growth?
- What are the total assets?

Limitations

- Only responds to exact predefined questions.
- Does not understand natural language variations.
- Uses hardcoded financial values.
- Cannot retrieve real-time financial data.
- Does not use machine learning or advanced NLP techniques.

Future Improvements

- Integrate pandas to read financial data directly from CSV files.
- Add Natural Language Processing (NLP) using libraries such as NLTK or spaCy.
- Create a web interface using Flask.
- Connect to live financial databases and APIs.
- Implement machine learning for smarter responses.